

Image Analysis Report

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Executive Summary

The image exhibits severe localized corrosion and advanced coating failure, particularly concentrated at a horizontal seam and a central hotspot. Given the heavy scale and exfoliation, this is classified as a high-severity condition requiring immediate intervention to prevent structural compromise.

Key Findings

#	Finding
1	Severe localized exfoliation and heavy iron oxide scale formation in the central region.
2	Preferential corrosion attack concentrated along the horizontal weld seam or lap joint.
3	Total coating breakdown in primary affected areas with active rust bleeding into surrounding regions.
4	Deep pitting visually apparent in the central zone, indicating probable significant wall loss.
5	Substrate roughness suggests the transition from atmospheric corrosion to localized material thinning.
6	Vertical rust staining suggests moisture runoff is exacerbating the corrosion process from the seam downwards.

Localized detections (VLM, approximate)

- Deep Pitting/Exfoliation (conf. 95%): x=0.340, y=0.120, w=0.380, h=0.720
- Corroded Seam (conf. 92%): x=0.000, y=0.450, w=1.000, h=0.120

Recommendations

#	Recommendation
1	Perform immediate Ultrasonic Thickness (UT) gauging to determine if the remaining wall thickness is within the engineering design limits.
2	Abrasive blast the affected area to Sa 2.5 standard to remove all corrosion products and allow for a detailed pit depth survey.
3	Conduct Non-Destructive Testing (NDT), such as Magnetic Particle Inspection (MPI), on the horizontal seam to check for corrosion-induced cracking.
4	If wall loss exceeds the specified corrosion allowance, perform a code-compliant weld overlay or section replacement.
5	Apply a high-performance protective coating system consisting of a zinc-rich primer and a high-build epoxy mid-coat.
6	Update the asset's risk-based inspection (RBI) profile to increase the monitoring frequency of this specific junction.

7	Investigate environmental factors, such as chemical exposure or moisture entrapment, that may have accelerated the degradation at this location.
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